

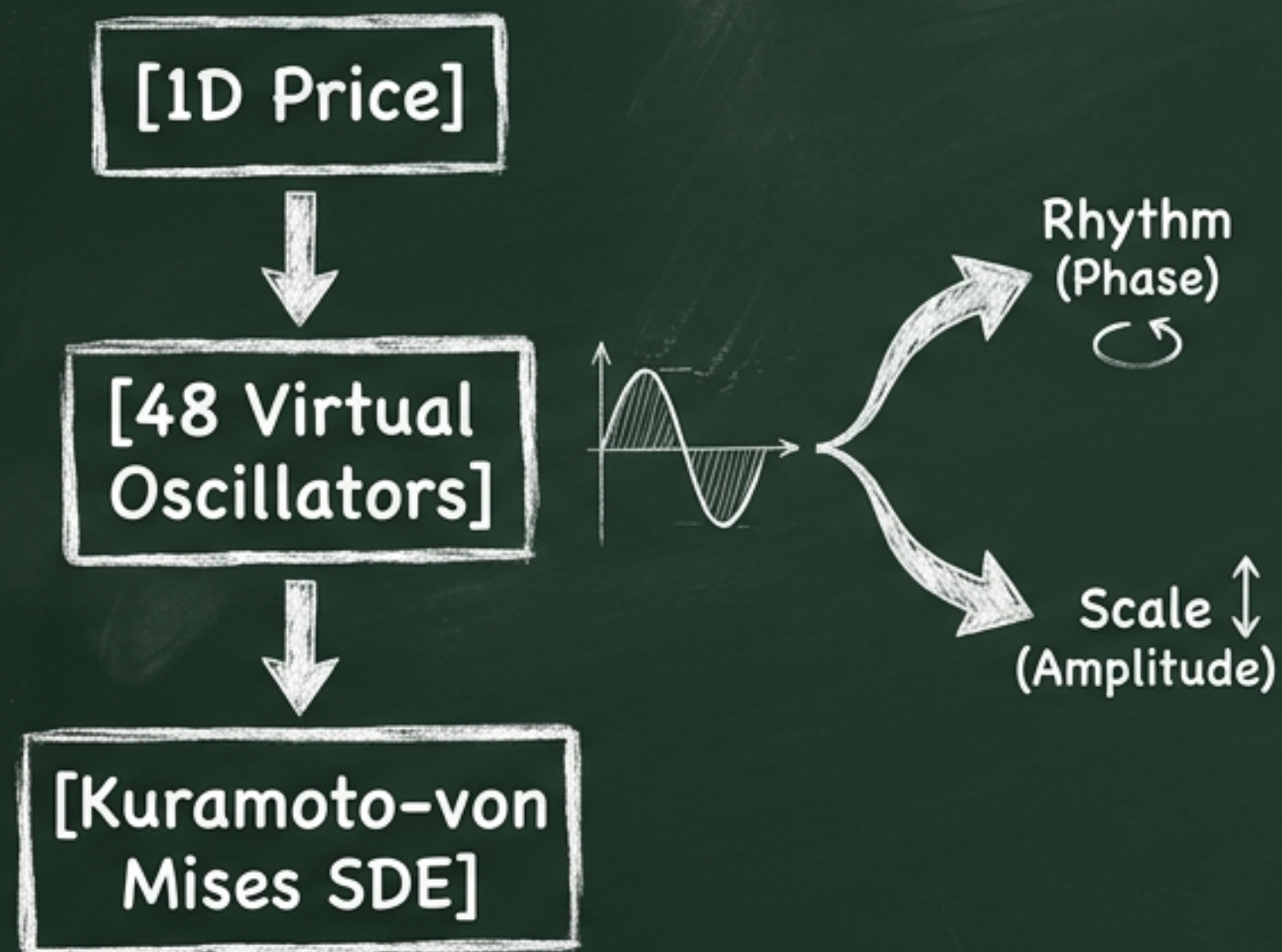
Restoring the Arrow of Time

How VMD Solves Spurious Coupling and Preserves Data Order in the PhaseSync SDE Model

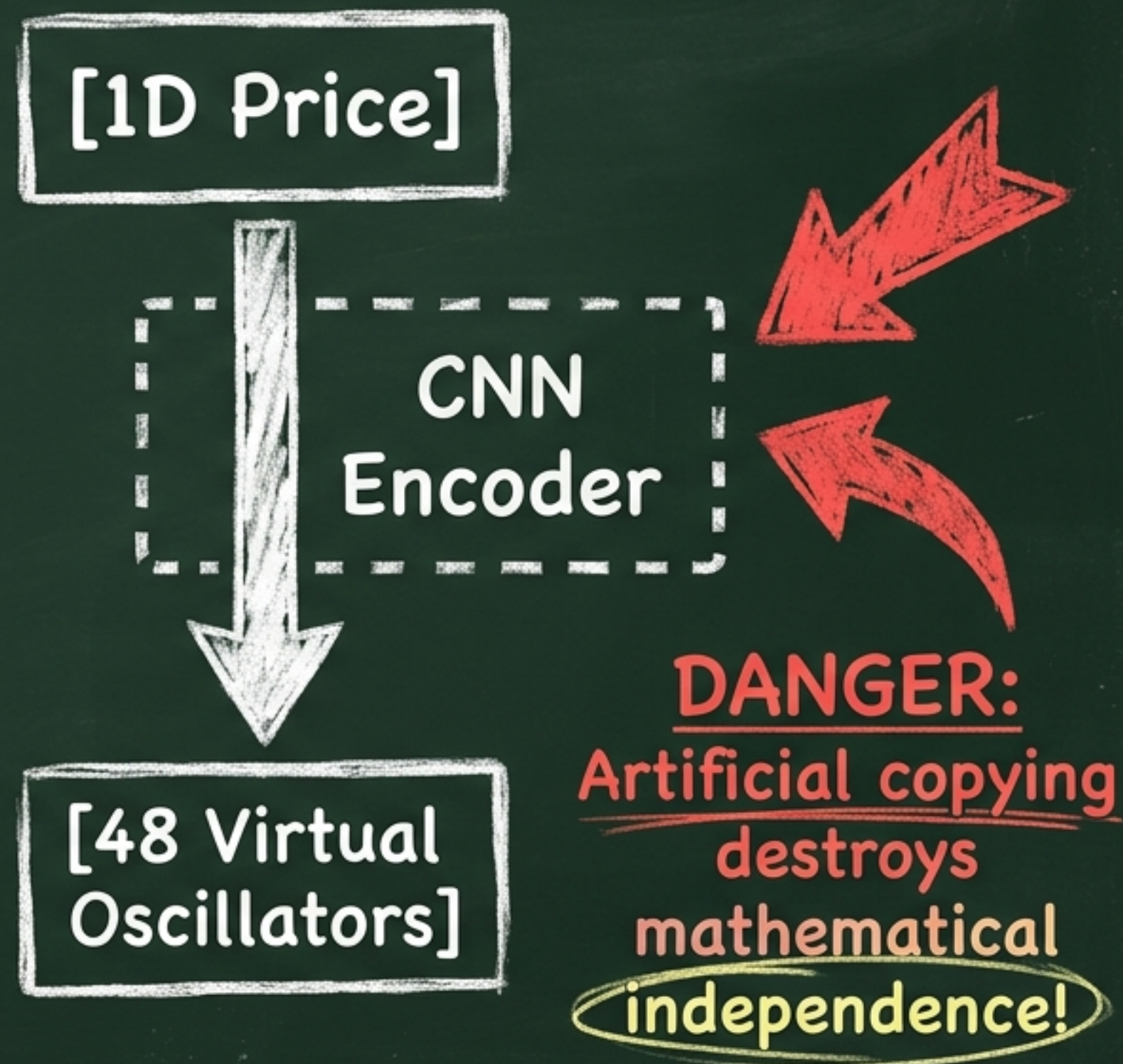


PhaseSync SDE Model Integration: Utilizing VMD-Separated IMFs for Accurate Causal Inference and Order Preservation.

The Goal: 1D Price $\rightarrow$ 48D Phase Space



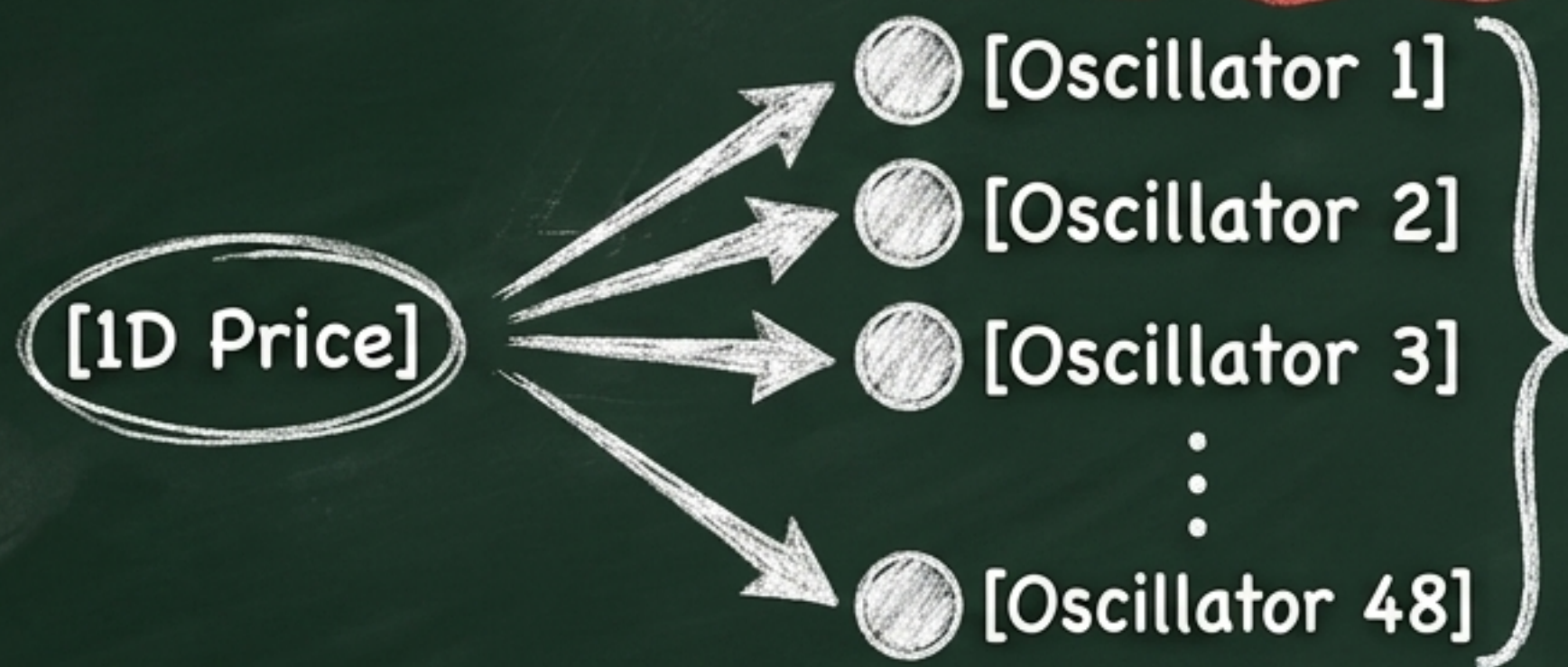
PhaseSync TSF models market turning points by simulating 48 "Virtual Oscillators" using non-equilibrium physics (KvM SDE).



The Problem: Breaking Spatial Conditional Independence

Anatomy of an Error

100% Correlated Noise



$$\hat{\mathbf{K}}_i = \cancel{M_i^{-1}} \mathbf{b}_i$$

The Math: Stanford's KvM closed-form solver requires conditionally independent noise across oscillators.

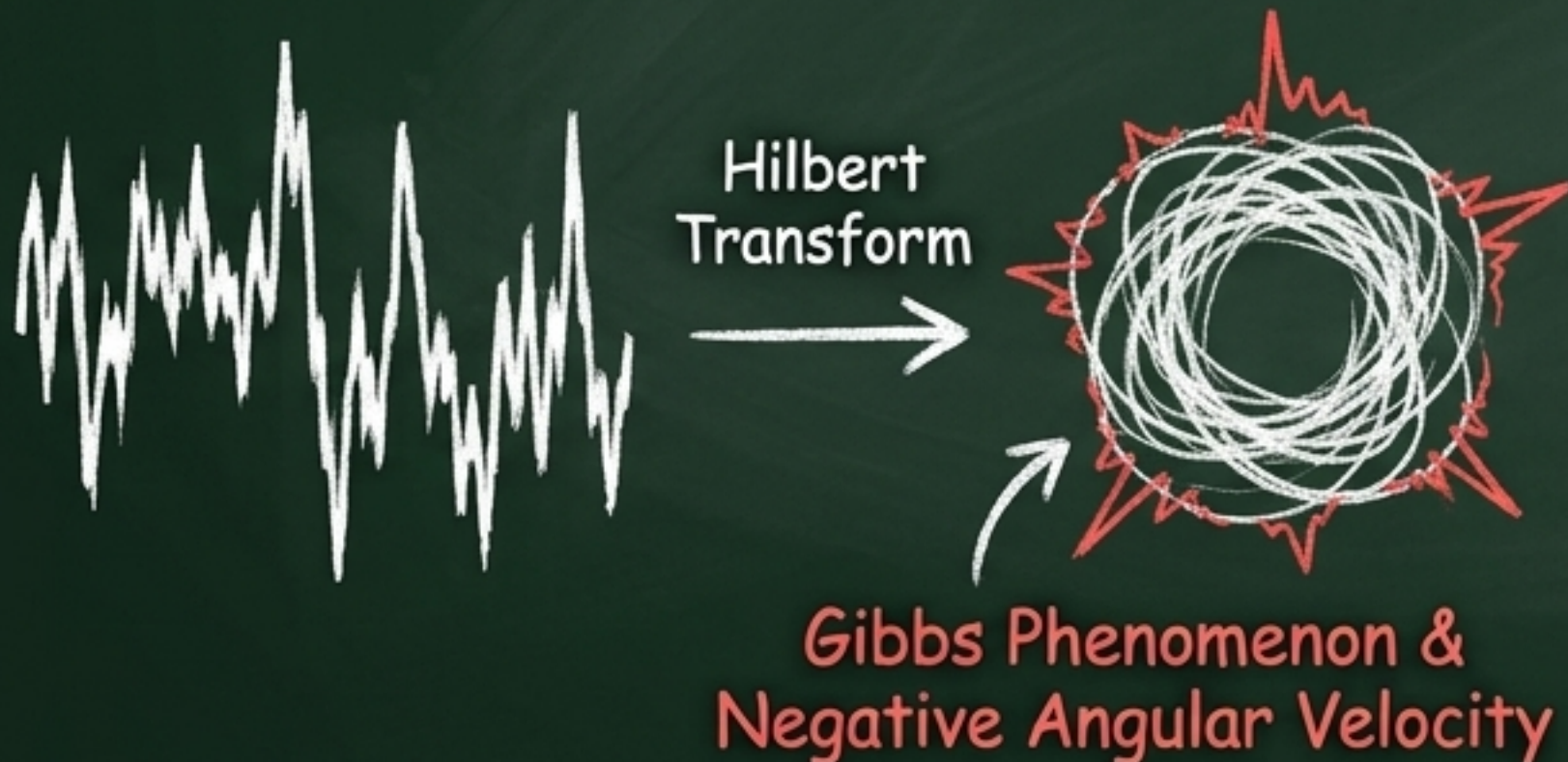
The Reality: Copying a 1D signal 48 times makes the underlying noise 100% correlated. The covariance matrix (M_i) suffers extreme multicollinearity.

Spurious Coupling: The model desperately overfits, creating massive, fake physical connections just to explain identical noise artifacts.

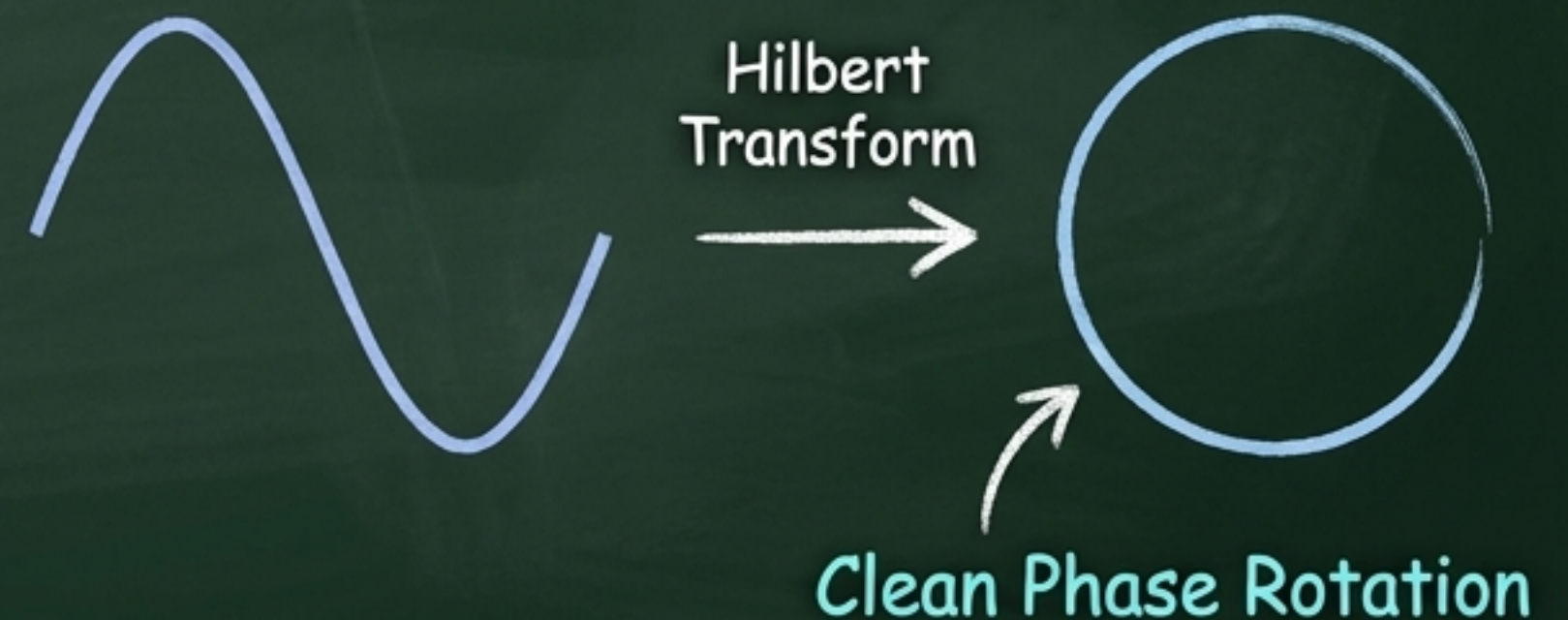
The Second Problem: Wideband Chaos

- Financial time series are wideband (daily noise + macro trends mixed).
- Forcing Hilbert transforms on raw price data breaks the chronological phase flow.
- KvM physics strictly require narrow-band inputs.

Wideband Chaos

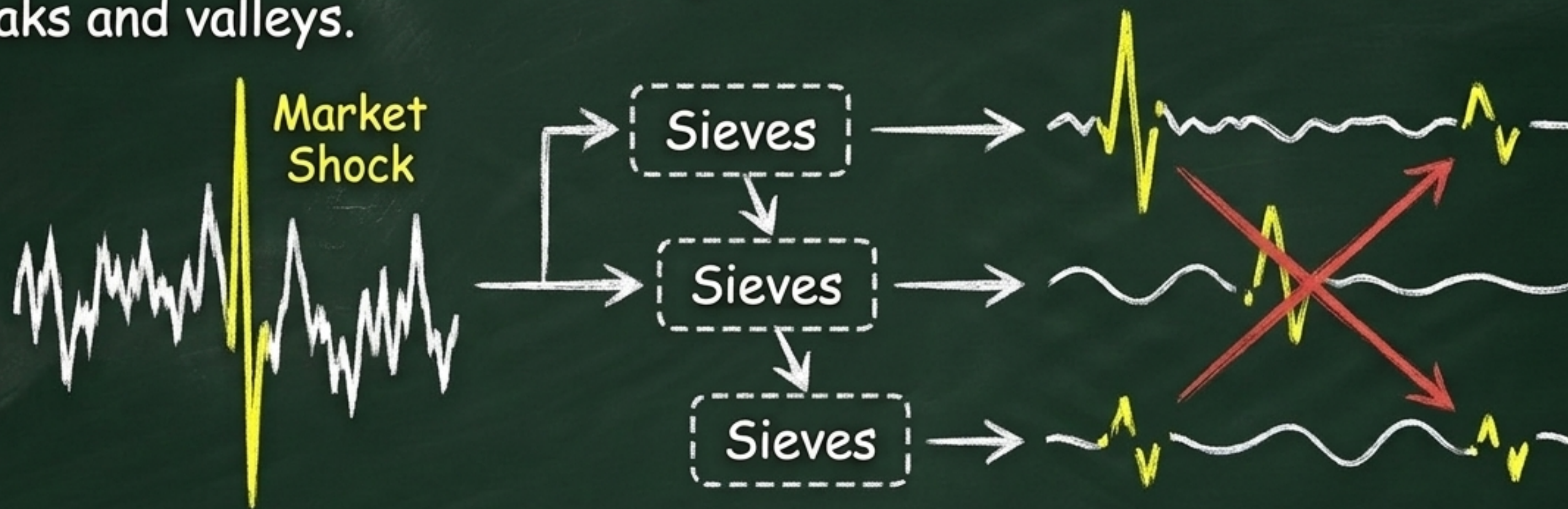


Ideal Narrow-band



Attempt 1: Empirical Mode Decomposition (EMD)

The Mechanic: EMD extracts signals sequentially based purely on local peaks and valleys.



The Fatal Flaw (Mode Mixing): When a sudden market shock occurs, EMD gets confused. Different frequencies bleed into the same mode.

Chronology Destroyed: A single real-world event is scrambled across multiple timeframes, irreparably breaking the causal sequence of the SDE simulation.

The Solution: Variational Mode Decomposition (VMD)

Analytic Signal: Strips negative frequencies.

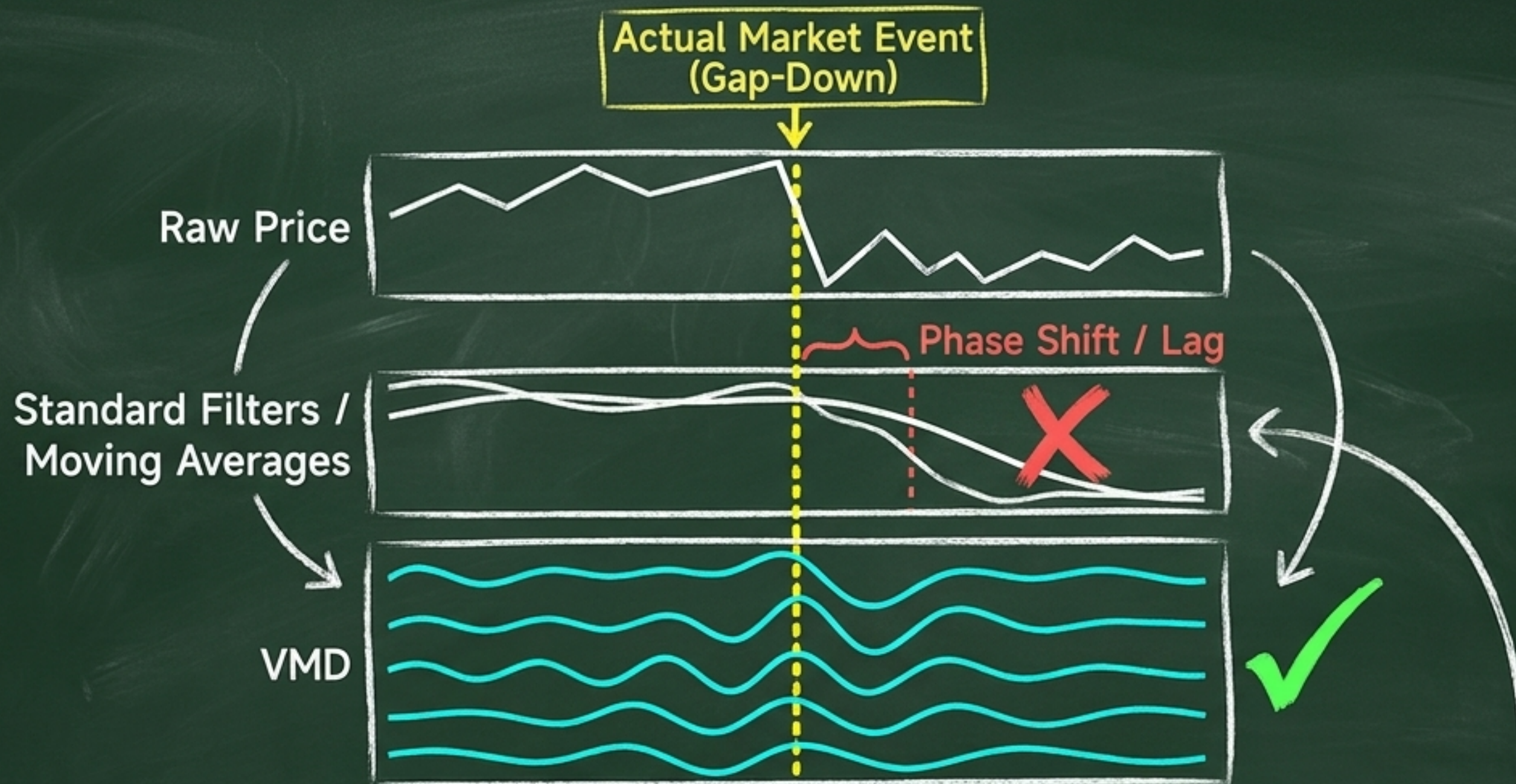
Baseband Shift: Pulls frequencies to the origin for perfect isolation.

$$\min_k \sum \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 \quad \text{subject to}$$
$$\sum u_k(t) = f(t)$$

L2 Norm Penalty: Forces extreme narrow-band smoothness.

Constraint: The ultimate data integrity check.

How VMD Maintains Chronological Data Order

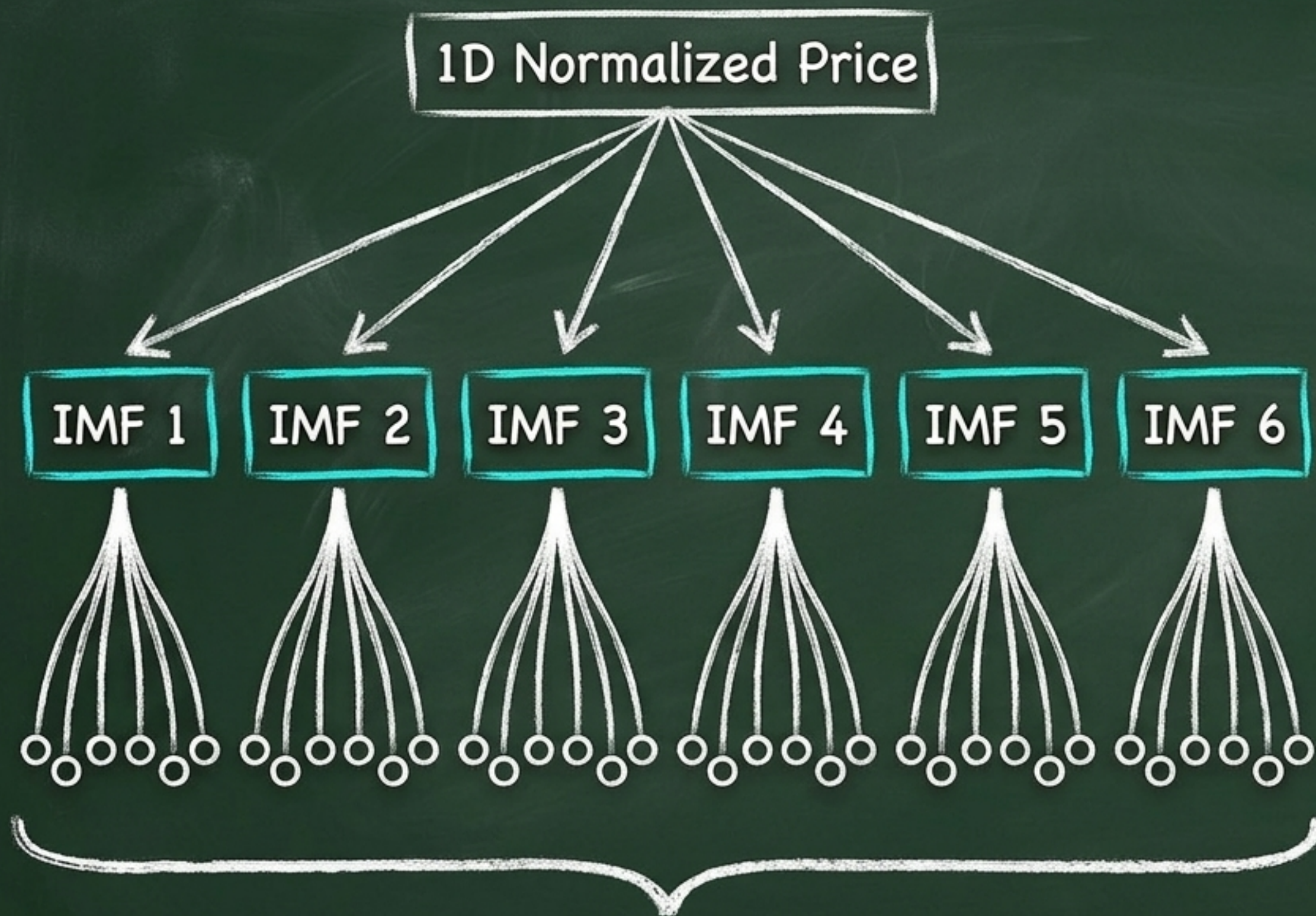


Simultaneous Optimization: VMD optimizes all modes globally in the frequency domain. It does not recursively distort the timeline.

Zero Phase Lag: Unlike rolling filters, VMD boundaries maintain strict chronological alignment with real-world events.

Perfect Reconstructability: The $\sum u_k = f(t)$ constraint mathematically guarantees no data point is delayed, lost, or leaked from the future.

Curing Spurious Coupling via Sub-Groups



6 IMFs $\times$ 8 = 48 Oscillators

- The Mechanic: Instead of copying the price, VMD splits it into 6 strictly orthogonal Narrow-band IMFs.

- The Mathematical Magic: Because VMD modes are orthogonal, the noise increments across the 6 groups are mathematically independent!

Result: Spatial Conditional Independence is perfectly restored.

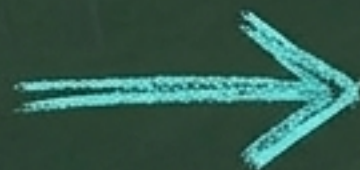
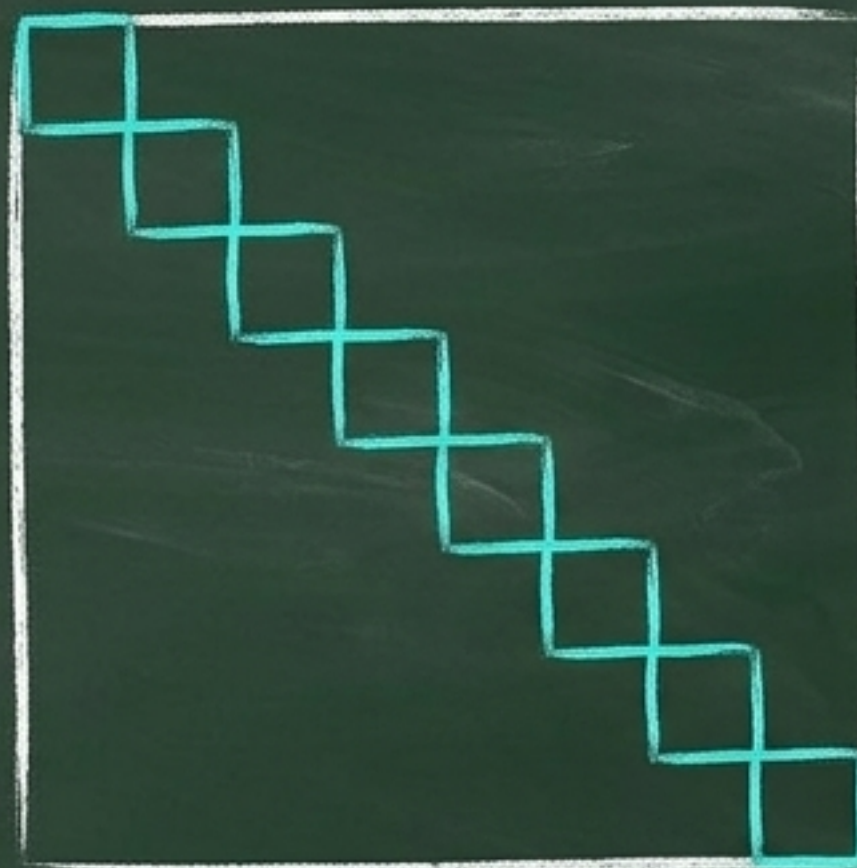
The "Stanford MLE" Works Again!

Covariance Matrix M_i

Before VMD



After VMD



Extreme Multicollinearity $\rightarrow M_i^{-1}$ Explodes

Orthogonal Modes $\rightarrow$ Stable Inversion

Key Insight: Because mathematical independence is restored, the ultra-fast linear algebra solver ($\hat{\kappa}_i = M_i^{-1} b_i$) finds the true physical lead-lag relationships of the market.

Result: Zero spurious overfitting. Pure, physical causality.

Connecting VMD to Heston Volatility



- The Problem: Traditional rolling windows create chronological lag.
- The VMD Advantage: Because VMD provides perfectly clean narrow-band IMFs, we apply the Hilbert Transform to extract instantaneous amplitude (volatility) in real-time.
- The SDE Integration: This instantaneous volatility directly drives the Heston CIR process, allowing the model to react to market shocks instantly without waiting for averages to catch up.

Architectural Diagnostic: Signal Decomposition for SDEs

Criteria	Naive CNN	EMD	VMD
Preserves Chronological Order?	✓	✗ (Mode Mixing)	✓ (Perfect)
Provides Independent Noise?	✗ (100% Correlated)	Partial	✓ (Orthogonal)
Creates Narrow-Band Signals?	✗ (Wideband)	Partial	✓ (Math Enforced)
Enables Closed-Form MLE?	✗ (Explodes)	✗ (Unstable)	✓ (Stable)

The Unified PhaseSync-VMD Pipeline



A perfect synthesis of non-equilibrium physics, precise signal processing, and deep learning—safeguarding the arrow of time from start to finish.